
Summary: Yamaura, Oka, Taki, Ozaki & Tanaka (2012), "Sustainable management of planted landscapes: lessons from Japan"

Source: *Biodiversity and Conservation* (2012) 21:3107-3129. DOI 10.1007/s10531-012-0357-4.

Core argument

Japan's planted-forest history is a global case study in both the risks of over-planting and the biodiversity value of *actively managing* (including harvesting) plantations, rather than either monoculture intensification or blanket preservation. The authors use Japan's trajectory to argue for a **hybrid "land sharing + land sparing"** strategy for planted landscapes worldwide.

Historical arc (mirrors Iwamoto/Totman but with an ecological lens)

- Grassland/coppice covered >10% of Japan through the Edo period, providing early-successional habitat for centuries; post-1950s, grasslands and coppice were converted almost entirely to conifer plantation (sugi, hinoki, larch) — now >40% of Japan's forest area, the world's 2nd-highest planted-forest share after Great Britain.
- Grasslands are now only 1-2% of land area. Their near-total loss is linked to a "second crisis of biodiversity" in Japan — the decline of early-successional species that depended on grassland/young-forest habitat, distinct from the "first crisis" (overexploitation of old-growth/natural forest).
- Most postwar plantations, now reaching harvest age, remain unharvested — owners are "reluctant to harvest," partly because replanting costs now often exceed timber revenue, turning forests into "long-term negative assets."

Quantitative demand modeling (their key policy number)

Assuming a 160-year maximum rotation and current age-class skew, at least **60,000 ha/year** must be cut on average just to keep pace — 1.7× the current cut rate — to meet even current domestic lumber/plywood demand. But because Japanese population is projected to fall to <40% of 2010 levels by 2110, and lumber/plywood demand has already been declining for decades, **only about 30% of existing planted-forest area is actually needed to meet future domestic demand**. The remaining ~70% could be allowed to revert toward natural forest, with or without active restoration.

Stand-level findings

- Monoculture sugi/hinoki plantations (dark, humid understory) support few bird species; larch plantations (deciduous, more light) support notably more biodiversity, especially old structurally complex stands.
- **Thinning** (especially heavy thinning) reliably increases broadleaf regeneration and species diversity even in dark evergreen conifer stands — a concrete, low-cost management lever.

- Retaining scattered natural/keystone trees (standing dead wood, remnant natural trees) within plantations supports significant biodiversity even under otherwise intensive management.

Landscape-level findings

- Forest-bird species richness is best explained by land use at 40-80km grid resolution, not finer scales; richness drops sharply once natural forest cover falls below **41%** of a region.
- Natural forest patches need to be large (>40 ha) to support high densities of many bird species — fragmentation into small patches undermines the sparing strategy.

National/international findings

- Japan's high wood demand met via imports has effectively **exported deforestation** to Southeast Asia (a major source of Japan's imported timber) — Japan has "underused its own forests and overused forests elsewhere." This is framed as a net biodiversity loss when counting Japan + its trade partners together.
- Migratory birds dependent on mature forest are declining in Japan (habitat loss abroad, in wintering grounds), while non-migratory mature-forest birds are increasing (Japan's own forests maturing) — a striking asymmetry illustrating the "export the impact" dynamic.

The land-sharing vs. land-sparing framework (central theoretical contribution)

- **Land sparing** = maximize yield on managed land, set aside large intact natural reserves elsewhere. **Land sharing** = lower-intensity, biodiversity-friendly management integrated across the whole landscape.
- Using Japan's own historical timber-price/biodiversity-value trajectory as a worked example, they show the "optimal" strategy has flipped over time: in 1966-1971 timber value dominated (favor production), by 1988-2003 the balance shifted toward biodiversity being undervalued relative to its true worth — implying Japan's management strategy should also have shifted, but largely hasn't.
- Their recommendation: a **hybrid** — retain some natural forest/patches intact throughout the plantation-management period (land sparing, for concave-response species and price-crash resilience), while managing the matrix itself (thinning, selective retention, eventual controlled clear-cut + partial natural regeneration) in a biodiversity-friendlier way (land sharing). This resembles the North American "TRIAD" model (intensive zone / extensively managed zone / strict reserve).
- Important practical point: clear-cutting mature plantations *without* replanting, followed by natural regeneration, can itself be a valid restoration pathway — recent studies found little erosion/landslide risk in abandoned clear-cut stands, and native canopy species can recolonize old plantations given time.

Relevance to Mitsue project

Directly actionable for [[project_reforestation_principle]] and the broader forest-workforce/CHP plan:

- Provides the ecological rationale (not just economic) for **harvesting rather than merely preserving** Mitsue's aging Sugi stands — under-harvesting is itself an ecological problem in Japan, reinforcing the CHP-fuel-supply thesis alongside [[reference_fit_grid_biomass]].
- The "only ~30% of planted forest needed to meet demand" finding is a strong national-level precedent for Mitsue's plan to redirect surplus/uneconomic Sugi stands toward biomass CHP fuel and let the rest

revert toward native broadleaf — validates the retention-plus-native-broadleaf approach already agreed with more trees ([[project_reforestation_principle]]).

- Thinning as a cheap, immediately actionable biodiversity lever is worth flagging to NGO/forestry partners (Niwa-san, [[project_sugano_fuel_partner]]) as a first-year, low-capex intervention that also produces chippable biomass.
- The land-sharing/land-sparing hybrid (TRIAD-like zoning) is a useful framing device for any funder-facing document explaining *why* Mitsue's plan combines active CHP-feedstock harvesting in some areas with native-forest restoration in others, rather than treating them as contradictory goals.